AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

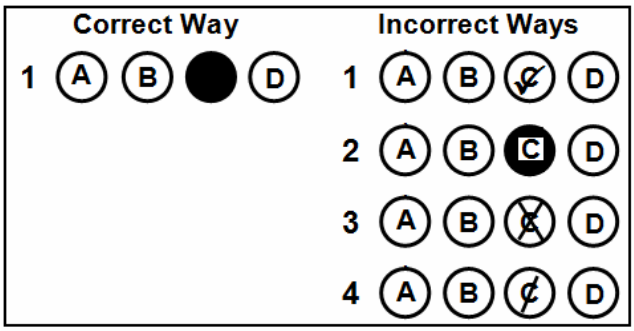
MAY 2013

Biology Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. Which of the following branches of biology deals with the study of mitosis and meiosis?

   A. Physiology
   B. Cell biology
   C. Taxonomy
   D. Histology

2. For an experiment, a biology student is given a potato cube. He has to find out its surface area to volume ratio.

   This case reveals the relationship of biology to

   A. chemistry.
   B. geography.
   C. economics.
   D. mathematics.

3. According to the scientific method of solving a biological problem, which of the following can not be opposed or rejected?

   A. Mass-flow hypothesis
   B. Mendel's laws of inheritance
   C. Darwin's theory of evolution
   D. A student's daily observations

4. Which of the following is a characteristic of a hypothesis?

   A. It should be a prediction of what will be observed.
   B. It should be based on the idea tested earlier.
   C. It should be written using complex words.
   D. It should be a very specific statement.

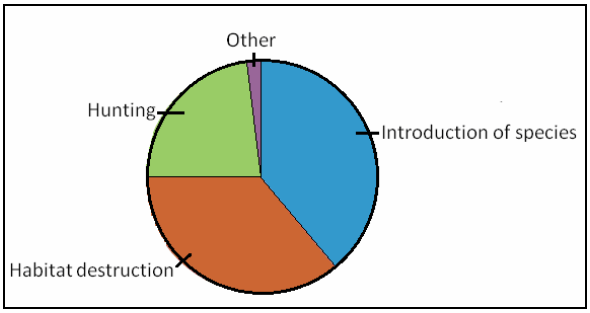
5. Which of the following is NOT a reason for the classification of living organisms?

   A. To highlight similarities among living organisms
   B. To bring to light differences among living organisms
   C. To count the total number of living organisms
   D. To explore evolutionary relationships among living organisms

6. If you have to divide five kingdoms of living organisms into two distinguishing groups, these can be

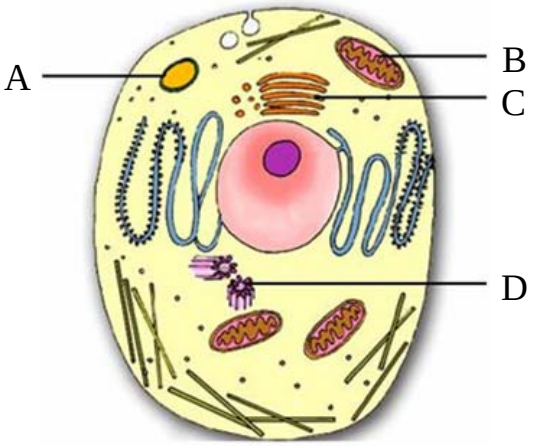
   A. prokaryotes and eukaryotes.
   B. autotrophs and heterotrophs.
   C. plants and animals.
   D. cellular and acellular.

7. According to the pie chart provided, which of the following measures is the most important to conserve the living organisms?



A. Climatic conditions should be stabilized.
B. Air pollution should be reduced.
C. Cutting of forests should be minimized.
D. Non-native species should be brought in.

8. Which of the labelled parts of an animal cell is responsible for converting molecules, coming from rough endoplasmic reticulum, into furnished products?



9. In the absence of which of the following organelles is a cell unable to breakdown food and waste material?

A. Golgi apparatus
B. Lysosomes
C. Endoplasmic reticulum
D. Ribosomes

10. Which of the following features of a leaf allows carbon dioxide to reach the inner cells of the leaf rapidly?

A. Thin Lamina
B. More chloroplast
C. Lignified xylem vessels
D. Large surface

11. Facilitated diffusion takes place through transport proteins. These transport proteins are present in the

A. cell wall.
B. cytoplasm.
C. cell membrane.
D. nuclear membrane.

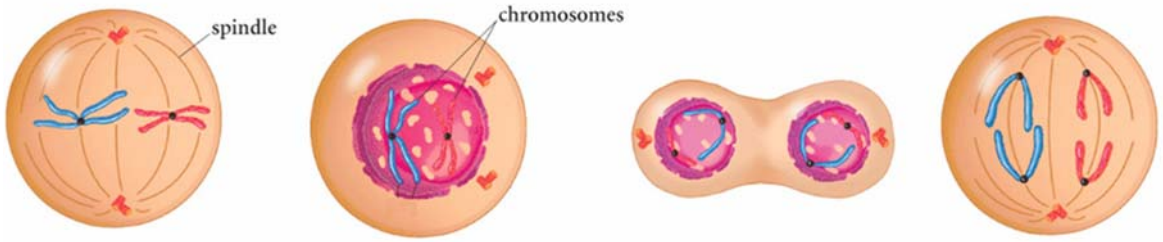
12. Which of the following is CORRECT about the cell cycle?

A. There are three main phases of the cell cycle.
B. S phase takes place before G1 phase.
C. It takes place in prokaryotic cells only.
D. Mitotic phase is shorter than interphase.

13. Skin cells do NOT enter G_o phase of the cell cycle as

A. they lack G1 phase.
B. they keep on dividing.
C. their supply of proteins is stopped.
D. their nuclei degenerate forever.

14. Which of the following diagrams shows metaphase of mitosis?

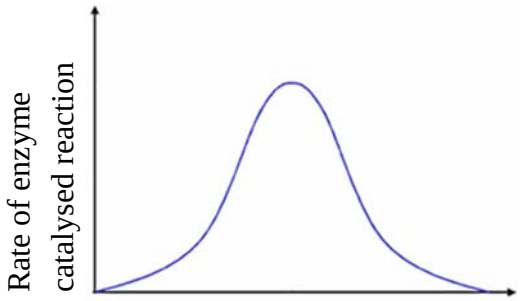


A B C D

15. Which of the following is CORRECT about the nature of chromosomes and pairing of chromatids during crossing over in meiosis?

	Nature of Chromosomes	Pairing of Chromatids
A.	Homologous	Non-sister
B.	Homologous	Sister
C.	Non-homologous	Non-sister
D.	Non-homologous	Sister

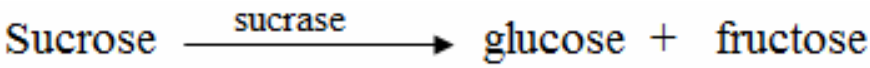
16. The diagram shows the effect of temperature on an enzyme catalysed reaction.



All of the following statements can be verified through the given graph EXCEPT that

A. enzymes can be denatured.
B. enzymes are protein in nature.
C. enzymes are temperature sensitive.
D. there is an optimum temperature for every enzyme.

17. In the given reaction, the active site is present in



A. sucrose.
B. sucrase.
C. glucose.
D. fructose.

18. When a molecule of adenosine diphosphate (ADP) reacts with water it gives

A. adenosine triphosphate and inorganic phosphate.
B. adenosine monophosphate and inorganic phosphate.
C. adenosine triphosphate and hydrogen gas.
D. adenosine monophosphate and hydrogen gas.

19. Which of the following chemical equations CORRECTLY shows the process of anaerobic respiration in man?

A. Glucose $\rightarrow$ Ethanol + CO_2 + Energy
B. Glucose $\rightarrow$ Ethanol + Energy
C. Glucose $\rightarrow$ Lactic acid + CO_2 + Energy
D. Glucose $\rightarrow$ Lactic acid + Energy

20. Which of the following may happen in a plant if it is grown without magnesium?

A. Increased water loss
B. Slow sugar transport
C. No chlorophyll formation
D. Deactivation of enzymes

21. A patient, besides taking basic nutrients, takes the following food items in his diet.

Carrots	Fish	Eggs



Through this diet, he is overcoming the deficiency of

A. vitamin E.
B. vitamin D.
C. vitamin C.
D. vitamin A.

22. Which of the following indicates the movement of larynx and state of trachea during the swallowing of food?

	Movement of Larynx	**Passage to Trachea**
A.	Up	Open
B.	Up	Closed
C.	Down	Open
D.	Down	Closed

23. In plants, water is carried upward by the xylem vessels through transpiration pull.

Which of the following transports water from the roots to the leaves?

A. Removal of water from the leaves
B. Repulsion between water molecules
C. Discontinuous cell wall
D. Large diameter of xylem vessels

24. Which of the following is CORRECT about eosinophils?

	Granular Cytoplasm	**Lobed Nucleus**	**Form in Lymph Nodes**
A.	✓	✓	×
B.	×	✓	✓
C.	✓	×	✓
D.	×	×	✓

25. In humans, the heart is considered as a ‘double pump’ because it

A. consists of auricles and ventricles.
B. circulates oxygenated and deoxygenated blood.
C. prevents backflow of blood by two valves.
D. possesses arteries and veins.

END OF PAPER

Please use this page for rough work